

ANALYTICAL GEOMETRY NOTES

Summary of Analytical Formulas:

a) Average Gradient:

The average gradient of a line passing through $K(x_K; y_K)$ and $L(x_L; y_L)$ is:

$$m_{KL} = \frac{y_K - y_L}{x_K - x_L}$$

b) Distance formula:

The distance between any two points $K(x_K; y_K)$ and $L(x_L; y_L)$ is:

$$KL = \sqrt{(x_K - x_L)^2 + (y_K - y_L)^2} \quad \text{or} \quad KL^2 = (x_K - x_L)^2 + (y_K - y_L)^2$$

c) Midpoint formula:

The coordinates of the midpoint of any line segment KL are:

$$\left(\frac{x_K + x_L}{2}; \frac{y_K + y_L}{2} \right)$$

d) Angle of inclination:

Angle of inclination is often shown as θ .

The gradient of a line (m) is equal to the tangent of the angle of inclination (θ).

$\tan \theta = m$ where $m \in (0^\circ; 180^\circ)$

e) Equation of a straight line:

$$y = mx + c \quad \text{or} \quad y - y_1 = m(x - x_1)$$

f) Equation of a horizontal line:

$$y = c \rightarrow c \text{ is a constant.}$$

g) Equation of a vertical line:

$$x = c \rightarrow c \text{ is a constant.}$$

h) Properties of parallel lines:

The gradients of parallel lines are equal: **$m_1 = m_2$**

i) Properties of perpendicular lines:

The product of the gradients of perpendicular lines is -1 :

$$m_1 \times m_2 = -1$$

j) Collinear points:

If points are collinear it means they lie on the same straight line.

If points A , B and C are collinear: **$m_{AB} = m_{BC}$** .

1. Finding the equation of a straight line:

Example 1:

Determine the equation of a straight which is:

- a) Parallel to the line $y = -3x + 4$; passing through the point A (4; 7).

$$y = -3x + c \rightarrow \text{line is } \parallel$$

$$\text{Sub A (4; 7)}$$

$$\therefore 7 = -3(4) + c$$

$$\therefore 19 = c$$

$$\therefore y = -3x + 17$$

- b) Perpendicular to the line $y = \frac{-2}{3}x + 2$; with a y -intercept of -3 .

$$y = \frac{3}{2}x + c \rightarrow \text{line is } \perp \text{ to } y = \frac{-2}{3}x + 2$$

$$\text{Sub } c = -3$$

$$\therefore y = \frac{3}{2}x - 3$$

- c) Parallel to the x -axis and passes through the point $(-4; 3)$.

$$y = 3 \quad (\text{A line parallel to the } x \text{ -axis is a horizontal line.})$$

- d) Parallel to the y -axis and passes through the point $(-4; 3)$.

$$x = -4 \quad (\text{A line parallel to the } y \text{ -axis is a vertical line.})$$

2. Angle of inclination of a straight line:

The angle of inclination is the angle measured anti-clockwise from the x -axis to a straight line. The angle of inclination is always between 0° and 180° .

Example: Determine the angle of inclination of the line $y = -4x + 2$.

$$m = -4$$

$\therefore \tan \theta = -4 \rightarrow$ shift $\tan^{-1} -4$ on your calculator. As the gradient is negative, the calculator gives an answer of $-75,96^\circ$. In order to obtain θ you have to add 180° .

$$\therefore \theta = 104,4^\circ.$$

3. Properties of quadrilaterals:

Example:

If a quadrilateral has vertices $A(-2; 2)$; $B(-3; -2)$; $C(3; -1)$ and $D(4; 3)$ prove that ABCD is a parallelogram.

$$M_{AC} = \left(\frac{-2+(3)}{2}; \frac{2+(-1)}{2} \right) \\ = \left(\frac{1}{2}; \frac{1}{2} \right)$$

$$M_{BD} = \left(\frac{-3+(4)}{2}; \frac{-2+(3)}{2} \right) \\ = \left(\frac{1}{2}; \frac{1}{2} \right)$$

$\therefore ABCD$ is a parallelogram (diagonals bisect)